SynthFlood: Ambient Synthetic Content Silently Erodes Web-Agent Decision Quality — A Dose–Response Analysis and a Provenance Remedy

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ABSTRACT

Web agents increasingly buy on our behalf, while an estimated quarter of the text they read is machine-generated under commercial optimization pressure. Unlike adversarial injection attacks, this *ambient* contamination targets no one — yet it may systematically distort what agents choose. We present **SynthFlood**, a paired, dose-controlled protocol that measures how synthetic product listings erode web-agent decision quality, built directly on WebShop's official data and reward function. Synthetic listings come from three families mirroring real content-farm production modes (search-optimized pages, specification clones with altered attributes and lower prices, review-inflated pages), generated by open-weight 7–8B models under a cross-model plausibility floor. Three findings emerge. First, a forced-choice probe (300 duels) shows agents buy near-correct specification clones at chance (52%), but controlled channel ablations show this is *not* a perceptual failure: normalizing the clone's price alone, or stripping its inflated ratings alone, each drop the clone win-rate to 39%; removing both drops it to 25% — below the 35% baseline for plain gloss — a clean two-channel decomposition. A dose–response over clone pricing shows the same clone is *rejected* at price parity (39%) but *preferred* under any discount of 10% or more (59% at −10%, $p = 0.003$): the attack does not fool the agent's perception, it buys the agent's preference. Second, contamination dose degrades decisions in the background: at $\varepsilon = 0.75$, task success drops 23 pp ($p = 1.3 \times 10^{-4}$, replicated across seeds) while regret rises 69% (paired $\Delta = +0.181$) and agents become *more* confident (calibration error doubling) — losses that no success-only metric would surface. Third, an agent-visible badge computed from a noisy verifiable-provenance bit eliminates the regret excess, remains unchanged from 15% to 65% badge forgery, and matches a perfect-provenance oracle when implemented as a *text-free structural* detector — a detector that fingerprint-free regeneration weakens only partially (28% of its recovery) while provenance stays intact; but the same information consumed as a soft ranking bonus fails until its weight exceeds the rating scale — provenance must order candidates, not score them. We release protocol, code, contamination pools, and full trajectory logs. These results recast ambient synthetic content as an attack on the agent's *preference function* rather than its perception, and call for provenance-ordered ranking as a first-class interface contract between the Web and the agents that act on it.

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1 INTRODUCTION

Web agents have crossed from demos to deployment: they shop, book, and compare on our behalf, turning the Web's product pages from artifacts that *inform humans* into signals that *direct machine decisions*. At the same time, the supply side of those pages is being rewritten. An estimated 27.5% of tokens passing common quality filters on the open Web are now AI-generated, and the share is climbing monthly [18]; measurement studies document machine-generated catalogs flooding music platforms [21] and scientific publishing [15]. This content is not adversarial in the classical sense: much of it is produced by ordinary commercial incentives — search-optimized content farms, counterfeit-style listing clones, review inflation — and a whole research program now studies how to *optimize* content for generative engines [1, 13, 22] and how to detect the result [5, 11, 28]. The inevitable consequence is that agentic systems increasingly browse a Web whose supply of plausible, keyword-rich, well-rated listings no longer correlates with actually satisfying the user's goal.

This paper asks a question that, to our knowledge, no prior work has measured in a controlled setting: **does ambient, non-targeted synthetic content silently erode the decision quality of web agents?** Existing evaluations of agents under Web pollution attack them deliberately: programmatic mini-internets with adversarially curated falsehoods [19], attacker-written endorsement pages steering search agents [4], or prompt injection and memory poisoning [3, 20, 23]. Upstream of the decision, retrieval-level studies show that cumulative AI pollution degrades search quality [26]. But ambient contamination is a different regime: no adversary knows the user's goal, so no page is crafted to defeat a specific decision. The threat is statistical, not surgical — and whether it materially matters for agent *economics* (choice quality, wasted spend, calibrated confidence) is unknown.

Measuring this regime requires solving a confound that has kept prior work at the retrieval layer: if we simply compare agent success before and after contaminating a benchmark, any degradation could

stem from task difficulty rather than choice quality. We therefore build **SynthFlood**, a paired, dose-controlled protocol on top of WebShop's official artifacts [24]. Each task renders a fixed candidate list in which the gold product is always present and never replaced; only distractor slots are progressively replaced by synthetic listings, drawn under a fixed seed so that lower contamination levels are nested within higher ones. Synthetic listings come from a three-family generator that mirrors real content-farm modes — search-optimized pages, specification clones with altered attributes and lower prices, and review-inflated plain pages — with a cross-model plausibility floor, all produced by open-weight 7–8B models. Agents act in a ReAct loop over rendered listings, and every purchase is scored by WebShop's *official* reward function, so regret is measured in the benchmark's own currency rather than a proxy.

SynthFlood yields three measurements. First, a *forced-choice susceptibility probe*: one gold product against one synthetic listing, 300 randomized duels, yielding a single interpretable number σ — the probability that an agent buys the machine-optimized page over the correct product when nothing else competes. Second, a *dose–response curve*: success, regret, and calibration as functions of the contamination share $\varepsilon \in \{0, 0.1, 0.25, 0.5, 0.75\}$, with McNemar tests on paired tasks. Our silent-erosion hypothesis predicts a characteristic signature here: regret rises steeply while coarse success appears stable, because partial mismatches (wrong attributes, overpayment) are invisible in success counts. Third, a *defense battery* evaluated at matched contamination, whose critical control is a random-defense arm that drops the same number of candidates uniformly — separating genuine filtering from mere candidate removal — alongside a provenance-only defense that re-ranks by a noisy verifiable-provenance bit without ever seeing labels.

Our contributions are:

- **A threat model and protocol** for ambient (non-targeted) synthetic contamination of agent decision channels, with paired dose–response control on official benchmark artifacts.
- **A forced-choice probe with channel decomposition** isolating *why* machine-optimized listings win: specification clones sit at chance ($\sigma_{\text{clone}} = 0.523$) while gloss and review inflation are rejected ($\sigma_{\text{geo}} = 0.351$, $\sigma_{\text{reviews}} = 0.287$); but price parity alone drops the clone win-rate to 0.386 (paired $p = 0.043$), stripping inflated ratings alone also yields 0.386, and removing both drops it to 0.250 — an additive two-channel decomposition showing the attack exploits price and social proof, not perception. A pricing dose–response yields a threshold law: any discount $\geq 10\%$ flips the majority to the synthetic clone ($0.386 \rightarrow 0.591$, $p = 0.003$), with no further gain at deeper discounts.
- **The first decision-level dose–response evidence** that ambient contamination erodes agent economics: at $\varepsilon = 0.75$, success drops by 23 pp ($p = 1.3\times10^{-4}$; $p = 1.2\times10^{-8}$ pooled over two seeds) and regret rises 69% (paired $\Delta = +0.181$, 95% CI $[+0.095, +0.264]$), while confidence rises and calibration error doubles — errors that no success metric would surface.
- **A defense analysis with an ordering principle**: an agent-visible noisy provenance badge eliminates the regret excess, is unchanged from 15% to 65% badge forgery, and a text-free structural detector statistically matches a perfect-provenance oracle ($p = 0.21$); but consuming the same bit as a soft ranking bonus fails until its weight exceeds the rating scale ($\lambda \leq 4$ all fail, $\lambda = 8$ recovers fully) — and a naive popularity ranking *backfires*, raising synthetic purchases to 67.5%.

We release all code, contamination pools, and full trajectory logs. The remainder of the paper formalizes the threat model (§3), presents the protocol (§4), reports results (§5), discusses implications and limitations (§6), and concludes (§7).

What is brave and new. This paper proposes that the Web's next decade will be shaped not by adversarial attacks on agents but by *ambient* commercial synthetic content — and that the resulting damage is invisible to the metrics we use to evaluate agents. What is new: (i) the first decision-level dose–response evidence that ambient contamination erodes agent economics on official benchmark artifacts; (ii) a channel-decomposed forced-choice probe that locates the erosion mechanism in the agent's *preference function* (price and social proof), not its perception, with a pricing threshold law showing a 10% discount is sufficient to flip the majority; (iii) a diagnostic triplet — flat success, rising regret, rising confidence — that operators can monitor; and (iv) the demonstration that a single noisy provenance bit recovers the full loss when consumed as an *ordering constraint* — but not as a soft score — where content-based detection cannot. What is brave: we argue provenance-ordered ranking should become a first-class interface contract between the Web and the agents acting on it, and we give the community the protocol to falsify or refine that claim.

2 RELATED WORK

AI-generated content on the Web. The scale of machine-written text on the live Web has moved from speculation to measurement: as of mid-2026, an estimated 27.5% of tokens passing common quality filters on the open Web are AI-generated, rising to 31.1% within two months, and high-quality human text is becoming scarce enough that its token value can be modeled with scaling laws [18]. Domain studies document the same shift operationally, from music streaming catalogs flooded with algorithmically produced tracks [21] to scientific publishing [15], and a growing conceptual literature debates what "slop" is and why it matters [10, 14]. Human-factors work shows that both lay readers [31] and automated pipelines struggle to distinguish such content. This literature treats synthetic content primarily as a *data-quality* problem for training corpora and a *perception* problem for readers. Our work complements it by asking a downstream question those studies do not touch: what happens when the consumers of this content are *autonomous agents* that act on it.

Generative engine optimization and content manipulation. A second line of work makes the production side explicit. GEO methods rewrite pages to maximize citation by generative engines [22], and a rapidly growing benchmark suite measures the resulting arms race: E-GEO formalizes seller-side optimization in e-commerce [1], GEO-Bench and its multimodal extension quantify ranking manipulation [6, 13], and IF-GEO studies multi-query optimization [29]. Detection responses are appearing at matching speed—GEO-Flag [5],

structural features that shape citation behavior [27], and Counter-GEO-Bench [28] together with defense-oriented work [11]. Two properties of this literature shape our contribution. First, manipulation is almost always *targeted*: an adversary crafts pages for specific queries or documents. Second, the affected system is the *engine*: what gets distorted is retrieval ranking or generated answers. We study the complementary regime—ambient, non-targeted contamination generated for commercial motives at scale—and the affected system is a *decision-making agent* that must commit to a purchase.

Agents under a polluted Web. Closest to us are systems that evaluate agents against corrupted information sources. The Synthetic Web builds programmatic mini-internets with adversarially curated pages and finds that language agents inherit epistemic weaknesses under single-injection falsehoods [19]. Endorsement-vulnerability measurements show that LLM search agents can be steered by attacker-published pages [4]. A separate security thread demonstrates prompt-injection and memory-poisoning attacks on web agents [3, 7, 20, 23], and persuasion studies show agent behavior can be shifted by conversational or inter-agent influence [8, 17, 30], with Sybil-style analysis questioning what counts as independent evidence [2]. At the retrieval layer, Yu et al. show that cumulative AI pollution eventually collapses search quality [26]. All of these works attack the agent with *deliberate, targeted* content, or measure degradation *upstream* of the decision. None asks the question we pose: whether *ordinary* optimization pressure—the same content farms that already dominate real e-commerce—degrades *decision quality* in a dose-dependent way, and what class of defenses restores it. To our knowledge, no prior work measures agent choice-level susceptibility with a forced-choice probe, a decision-level dose–response function, and regret-based economics on a grounded commerce benchmark.

Shopping-agent benchmarks. WebShop [24] established grounded language agents over real e-commerce data with attribute-level rewards; successors extend it to personalized curation [9], multi-shop comparison [16], and deep research shopping [12]. ReAct-style prompting [25] remains the standard operating mode for these agents. We reuse WebShop's official data and reward function—unmodified—as the substrate for a controlled candidate-selection environment, which lets us attribute every reward difference to the manipulated variable (which listing the agent chose) rather than to benchmark artifacts.

3 PROBLEM FORMULATION

3.1 Setting: ambient contamination of a decision channel

Consider an agent A that, given a user instruction u, must choose one product from a candidate list $C = \{c_1, \ldots, c_N\}$ rendered by a search or recommendation layer. Each candidate is a *listing*: a title, marketing bullets, specification attributes, a description, a price, and an aggregate rating with a review count. The agent observes a rendered version of these fields and commits to a choice $k \in \{1..N\}$.

We model *ambient synthetic contamination* as a random process on the candidate pool, not as an adversary. A fraction ε of distractor slots is occupied by synthetic listings s produced by a generative model under *commercial* optimization pressure: maximize apparent fit to the user's intent (query keywords, bullet structure, rating and review volume), minimize effort. Formally, for a task with gold product g and real distractors $D = \{d_1, \ldots, d_{N-1}\}$, the contaminated pool is $C_\varepsilon = \{g\} \cup D' \cup S$, where S replaces $\lfloor \varepsilon(N-1) \rfloor$ elements of D. The gold product is never replaced and the positions of all candidates are fixed per task, so that measurements at different ε are *paired*.

Three properties distinguish this threat model from adversarial injection. (i) *No targeting*: synthetic listings are produced for the category and query in general, not against a specific user, task, or gold product; their accidental overlap with any given instruction is incidental. (ii) *No intentional falsehood*: the generator is not told the gold attributes it must misrepresent; it produces plausible-looking, keyword-rich pages whose specifications may or may not match any particular user's requirements. (iii) *Persistence*: such content is the by-product of ordinary incentives (SEO, affiliate monetization), so its prevalence ε is a property of the Web, not of an attack session.

3.2 Quantities of interest

Let $r(c; u) \in [0, 1]$ be the official reward of candidate c under instruction u, and let $K_\varepsilon \in \{1..N\}$ be the agent's choice at contamination level ε. We measure:

- **Task success** $\mathrm{S}(\varepsilon) = \Pr[\, r(c_{K_\varepsilon}; u) \geq 1 - \delta \,]$ with $\delta = 10^{-3}$ (and a partial-success variant at $\delta = 0.2$), and **regret** $\mathrm{R}(\varepsilon) = \mathbb{E}[\, 1 - r(c_{K_\varepsilon}; u) \,]$, the economic loss of acting on the contaminated Web relative to the best available candidate.
- **Susceptibility** σ, from a forced-choice probe: with only two candidates, the gold g and one synthetic listing s, presented in randomized order, $\sigma = \Pr[K = s]$; the null value is $\sigma = \frac{1}{2}$, and $\sigma > \frac{1}{2}$ indicates a systematic preference for machine-optimized pages independent of any dose–response accumulation.
- **Dose–response curve** $\mathrm{S}(\varepsilon)$ over the grid $\varepsilon \in \{0, 0.1, 0.25, 0.5, 0.75\}$, summarized by ε_{50}, the contamination level at which clean-condition success halves.
- **Calibration drift**: expected calibration error (ECE) of the agent's post-hoc choice confidence as a function of ε, capturing whether contamination erodes *confidence* even when choices are still correct.

3.3 The silent-erosion hypothesis

The security literature on web agents predicts degradation only when adversarially targeted content is present. Ambient contamination implies a different, more insidious signature: because synthetic listings are optimized for *perceived* quality rather than goal satisfaction, we expect the choice distribution to shift toward them ($\sigma > 1/2$; R increasing in ε) while coarse success rates may appear stable — regret rises through partial-quality purchases (attribute mismatches, overpayment) long before complete failures appear. A dose–response with this shape would mean the Web can lose decision quality at a nearly invisible rate to the operators of agents, motivating provenance-ordered defenses rather than detection alone. Our experiments are designed to falsify or confirm this signature (Section 5).

4 THE SYNTHFLOOD PROTOCOL

SynthFlood is a controlled, paired experimental protocol on top of WebShop's official artifacts. It consists of a candidate-selection environment (§4.1), a three-family contamination generator with a plausibility floor (§4.2), a forced-choice susceptibility probe (§4.3), a dose–response sweep with matched defenses (§4.4), and a gated execution discipline that fixes the analysis before scaling compute (§4.6).

4.1 WebShop-Select: a controlled selection environment

We load WebShop's official product dump (1,181,430 products after the official deduplication), the official attribute file, and the official human instruction set, using a verbatim port of the benchmark's loader; goals are produced by the benchmark's own `get_human_goals`, including its price-upper sampling. Rewards are computed by the benchmark's official reward function (attribute matching over specifications, title, bullets and description; option matching; price constraint; type gate), applied to the chosen candidate. Because our protocol removes the variant-selection step, we score the chosen listing with its best-matching variant, which keeps the option term comparable across candidates: all reward differences then stem exclusively from *which listing* was chosen, the manipulated variable.

Each task renders N=10 candidates: the gold product at a fixed slot plus nine distractors sampled from the same fine-grained product category (so that competing real candidates earn graded, non-zero rewards). The agent observes a textual results page and may issue `view[k]` to inspect a full product page or `select[k]` to purchase; episodes are capped at ten steps. The gold product is always present and never replaced. At contamination level ε, $\lfloor \varepsilon(N-1) \rfloor$ distractor slots are replaced by synthetic listings drawn from a per-task pool with a fixed seed, so the candidate list is *identical across ε levels except for the replaced slots* — a paired design that isolates the effect of contamination from task difficulty.

4.2 Contamination generator with a plausibility floor

Three families model the dominant production modes of commercial synthetic content. **geo**: a search-optimized content-farm page that targets the query keywords with a polished title, marketing bullets, and specification lines, generated *without access to the gold attributes*, so its specifications violate one or two of the instruction's required attributes with high probability. **clone**: a near-copy of the gold listing with one or two critical attributes altered and a 5–25% lower price, counterfeit-listing style. **reviews**: an otherwise plain page flooded with twenty generic five-star machine-written reviews. All listings receive programmatic economics (rating 4.6–4.9, 400–3,000 reviews), inherit the gold listing's option schema (which enters only the reward's option term), and must pass a *plausibility floor*: a judge model, distinct from the generator, scores each listing 1–5 for plausibility as a real e-commerce page, and listings below 4 are regenerated once and dropped if still below. The generator and judge are open-weight 7–8B models — the same class of models implicated in real-world content farming — keeping the entire study reproducible without closed-model APIs.

4.3 Forced-choice susceptibility probe

Dose–response curves confound preference strength with slot competition. We therefore run an existence-first probe: each task presents *exactly two* candidates — the gold product and one synthetic listing, order randomized by a fixed seed — and the agent must buy one. The outcome is a Bernoulli choice; across 30 tasks × 10 pairs = 300 forced choices, susceptibility $\sigma = \Pr[K = s]$ is estimated with a Wilson 95% interval, and the null $\sigma \le \frac{1}{2}$ is tested binomially. This probe requires no reward model interpretation, is robust to task difficulty heterogeneity (each task contributes its own pairs), and yields an interpretable headline number: the probability that an agent buys the machine-optimized page when it competes one-on-one with the correct product.

4.4 Dose–response sweep and defense battery

The main measurement sweeps ε over $\{0, 0.1, 0.25, 0.5, 0.75\}$ under `mix` contamination (families drawn uniformly), reporting success, partial success, regret, ECE, and per-family breakdowns, with McNemar tests between adjacent doses on paired tasks and cluster bootstrap confidence intervals over tasks. The defense battery runs at a fixed high contamination level and compares seven conditions: **none** (vanilla); **prompt** (a standing instruction to distrust keyword-rich, specification-mismatched pages); **detector-text** (perplexity filtering); **detector-struct** (structural slop features: marketing cue density, keyword stuffing, bullet symmetry); **random** (the same number of candidates dropped uniformly at random — the critical control that separates *filtering quality* from *merely removing candidates*); **provenance** (a noisy provenance bit: real items carry verifiable provenance 85% of the time, synthetic 15%, deterministic per item; the defense re-ranks by this bit alone, never seeing labels); **oracle** (perfect re-ranking, an upper bound). Defenses are evaluated on regret recovery relative to none at matched contamination.

4.5 Channel ablations and defense stress tests

To attribute the probe's clone result to specific listing channels, we re-run the forced-choice protocol with surgical edits to the synthetic clone only: **price parity** (clone re-priced exactly at the gold price), **rating parity** (rating and review count reset to the gold listing's genuine values), and **both**; a fourth dose series fixes the clone's price at a fraction $f \in \{0.9, 0.8, 0.7\}$ of the gold price. Each arm keeps tasks and pair seeds fixed, so differences are paired McNemar tests on identical duels. On the defense side we add three stress tests: a **soft-ordering sweep** re-implementing the provenance bit as an additive bonus $\lambda \cdot \mathbb{1}[\text{verified}]$, $\lambda \in \{0.5, \ldots, 8\}$; a **forgery envelope** letting the adversary forge the badge on $s \in \{25, 35, 50, 65\}\%$ of synthetic listings; and a **naive popularity baseline** ranking by rating× log(review count). Finally, to probe the structural detector's reliance on generator fingerprints, a **stealth pool** is regenerated with fingerprint-free prompts (no marketing adjectives, naturally varied bullet lengths, prose-form specifications, humanlike reviews) and the defense battery re-run on it.

4.6 Gated execution

All stages are run under a pre-registered gate discipline: P0 (pipeline sanity: zero malformed actions, success in a plausible range, seed

reproducibility) precedes generation; the generator passes a plausibility and manual audit; P1 gates on $\sigma \geq 0.60$ (binomial $p < 0.01$ over 300 choices) — if agents show no systematic preference for synthetic listings, the premise is falsified and the project pivots; P2 gates on a ≥5 pp success drop or ≥10% relative regret increase at $\varepsilon = 0.75$; a power analysis then fixes the final matrix before the full sweep; P5 gates on any defense recovering ≥30% of the incurred regret. Gate decisions and full trajectory logs (JSONL, every LLM call with token counts) are released with the code.

5 RESULTS

5.1 Forced-choice probe: clones win, but why?

The forced-choice probe confronts each agent with one gold product and one synthetic listing (300 randomized duels). Agents overall still prefer the real product, $\sigma = 0.377$ (95% CI [0.324, 0.433], binomial vs. $\frac{1}{2}$ $p < 10^{-35}$), but the families separate sharply: $\sigma_{\text{clone}} = 0.523$ — statistically indistinguishable from chance — while $\sigma_{\text{geo}} = 0.351$ and $\sigma_{\text{reviews}} = 0.287$. Taken alone, this pattern invites a perceptual reading: specification clones seem to pass as real products. Trajectory inspection is consistent with such a reading — agents quote instruction attributes and buy the real product even when it displays no rating — but trajectories cannot separate what the agent *fails to see* from what it *prefers to buy*. The next subsection closes that gap with controlled channel ablations, and reaches the opposite conclusion.

5.2 Channel decomposition: preference, not perception

We re-run the clone duels under three controlled ablations of the synthetic listing: **price parity** (clone priced exactly at the gold price), **rating parity** (rating and review count reset to the gold listing's genuine values), and **both** (300 duels each, same tasks and pair seeds). Table 1 gives the result. Either channel alone drops the clone win-rate from 0.523 to 0.386 (price parity: paired McNemar $+9/-21$, $p = 0.043$; rating parity: $+5/-17$, $p = 0.017$); removing both drops it to 0.250 (95% CI [0.171, 0.350]; paired $+3/-27$, $p < 10^{-4}$). The decomposition is additive to within half a point: $0.523 - 0.137$ (price) -0.136 (social proof) $= 0.250$ (Figure 1a). Two conclusions follow. First, the clone's advantage is *bought* with a discount and manufactured ratings, in equal shares — not earned by perceptual similarity: at full price and honest ratings the clone is rejected almost twice as often as chosen. Second, the residual 0.250 lies *below* the plain-gloss baseline $\sigma_{\text{geo}} = 0.351$: once the economic lures are removed, a listing that near-copies the gold product's specifications is a *liability* for the attacker, plausibly because the close-but-altered specification text makes the mismatch easier to notice.

A threshold law for discount pricing. Because price is the dominant channel, we dose it directly: the clone is priced at a fixed fraction $f \in \{1.0, 0.9, 0.8, 0.7\}$ of the gold price with ratings left inflated (300 duels per level; paired against the $f = 1.0$ arm). The response is a *threshold*, not a gradient: at price parity agents reject the clone ($\sigma = 0.386$), at −10% they already prefer it ($\sigma = 0.591$; paired $+26/-8$, $p = 0.003$), and deeper discounts add nothing (0.534 at −20%, $p = 0.015$; 0.568 at −30%, $p = 0.004$) — all discount levels sit within noise of the original mixed-discount pool (0.523). Figure 1b plots the response. In deployment terms, *any* meaningful discount suffices for the attack; the agent's price sensitivity saturates immediately past the threshold.

Table 1: Channel decomposition of the clone win-rate σ (gold vs. synthetic clone, 300 duels per row; paired flips vs. the unmodified clone).

Clone variant	σ	95% CI	Paired vs. orig.
original (price −5 to −25%, inflated)	0.523	[0.420, 0.624]	—
price = gold	0.386	[0.291, 0.491]	$+9/-21$, $p = 0.043$
rating = gold	0.386	[0.291, 0.491]	$+5/-17$, $p = 0.017$
both (full parity)	0.250	[0.171, 0.350]	$+3/-27$, $p < 10^{-4}$
gloss baseline (geo family)	0.351	[0.263, 0.450]	—

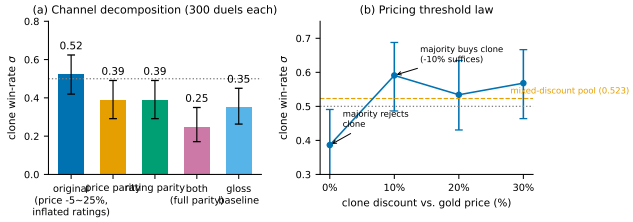


Figure 1: The erosion mechanism is preferential, not perceptual. (a) Channel decomposition of the clone win-rate: price parity and rating parity each drop σ from 0.523 to 0.386; full parity drops it to 0.250, below the plain-gloss baseline. (b) Pricing threshold law: the same clone is rejected at price parity but majority-preferred at any discount ≥ 10%; deeper discounts saturate near the mixed-discount pool level. Error bars: Wilson 95% CIs; dotted line: chance.

5.3 Dose–response: success flat, regret and confidence climb

Figure 2 and Table 2 give the main curve over the effective dose grid $\varepsilon \in \{0, 0.25, 0.5, 0.75\}$ (with N=10 candidates, the nominal 0.1 dose maps to zero replaced slots and is omitted; each dose corresponds to 0, 2, 4, or 6 replaced slots), N=100 tasks per dose under the Qwen2.5-7B backbone, plus a full second-seed replication of the endpoints (N=200 paired observations). Three co-movements define the silent-erosion signature. **Success collapses quietly**: from 0.590 at $\varepsilon = 0$ to 0.360 at $\varepsilon = 0.75$ (McNemar on paired tasks: 28 vs. 5 discordant, $\chi^2 = 14.7$, $p = 1.3 \times 10^{-4}$; pooling both seeds, 64 vs. 13, $\chi^2 = 32.5$, $p = 1.2 \times 10^{-8}$). **Regret climbs even faster**: mean regret 0.262 → 0.443 (+69%), paired endpoint difference +0.181 (bootstrap 95% CI [+0.095, +0.264], $t = 4.2$). **Confidence rises throughout**: post-hoc verbalized confidence increases from 0.822 to 0.875 while expected calibration error *doubles* (0.249 → 0.496) — agents err more while trusting themselves more. The purchase share tells the mechanism directly: synthetic listings are bought in 53%, 42%, and 54% of tasks at $\varepsilon = 0.25, 0.5, 0.75$ respectively (from 0% at $\varepsilon = 0$; Cochran–Armitage trend $z = 6.98$, $p = 1.4 \times 10^{-12}$),

Table 2: Dose–response of agent decision quality under ambient contamination (Qwen2.5-7B backbone, mix contamination, N=100 tasks per dose; P4A). Confidence is the mean post-purchase verbalized confidence; ECE is expected calibration error over five equal-width bins.

ε	Success	S@0.8	Regret	Synth. bought	Conf.	ECE
0.00	0.590	0.600	0.262	0%	0.822	0.249
0.25	0.350	0.350	0.429	53%	0.866	0.498
0.50	0.420	0.420	0.401	42%	0.882	0.453
0.75	0.360	0.360	0.443	54%	0.875	0.496

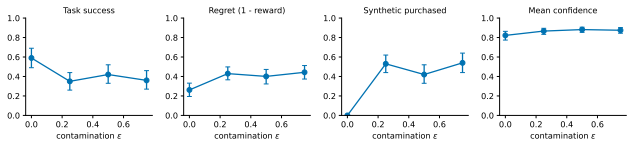


Figure 2: Dose–response under ambient contamination: task success, regret, synthetic-purchase share, and mean confidence as functions of the contamination level ε (mix families; N=100 tasks per dose; error bars = 95% cluster-bootstrap CIs over tasks).

Table 3: Per-family contamination at $\varepsilon = 0.75$ (40 paired tasks).

Family	Success_0	$\text{Success}_{.75}$	Regret_0	$\text{Regret}_{.75}$
geo	0.575	0.575	0.296	0.331
clone	0.600	0.425	0.273	0.384
reviews	0.575	0.425	0.294	0.454

and purchases of synthetic listings carry mean reward 0.41 versus 0.79 for purchases of real listings at the same dose.

5.4 Family attribution

Single-family runs at $\varepsilon = 0.75$ (Table 3) attribute the damage. Clone contamination is the most destructive at the success level (success 0.600 → 0.425, −17.5 pp; regret +41%; synthetic listings bought in 52.5% of tasks), consistent with the probe's decomposition — clones carry the two lures the agent is most sensitive to. Review-inflated pages deliver the largest regret shock (0.294 → 0.454, +54%): a plain page with a manufactured five-star flood gets purchased (37.5% of tasks) precisely because it looks credible, then fails on missing specifications. Geo contamination is the mildest — success flat, regret +11.8% at seed 1 — but we flag the attribution honestly: a full second-seed replication of the geo family gives a pooled paired regret difference of +0.075 whose 95% CI [−0.050, +0.195] crosses zero, so we treat "quiet attrition" for this family as *directional* evidence only. Two facts about geo contamination are nevertheless solid: it does not measurably change task success, and when a geo page *is* purchased it earns a mean reward of only 0.22 (versus 0.985 for gold purchases in the same arm) — the rare geo purchase is a near-total loss.

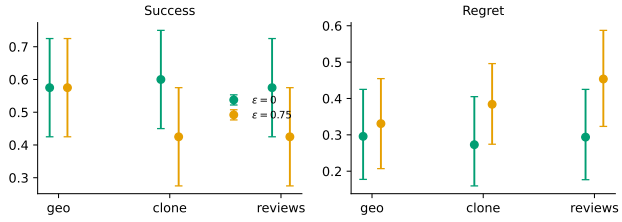


Figure 3: Per-family contamination at $\varepsilon = 0.75$ versus the paired uncontaminated baseline (40 tasks): success (left) and regret (right). Specification clones destroy success; review-inflated pages inflict the largest regret; search-optimized pages cause quiet attrition.

5.5 Defenses: provenance beats detection

Table 4 evaluates the defense battery at $\varepsilon = 0.75$ against the paired uncontaminated baseline (same tasks, same seeds; the none row is the unprotected high-dose condition). Provenance re-ranking with an agent-visible `[verified]` badge — computed from a *noisy* provenance bit (85% coverage on real items, 15% spurious on synthetic ones) — recovers the full regret excess (0.398→0.329, versus the uncontaminated 0.338) and cuts synthetic purchases from 47.5% to 32.5%; the oracle badge (perfect provenance) restores success *above* the uncontaminated level (0.750 vs. 0.525) and pushes regret below it (0.171), because knowing which listings are verified makes agents scrutinize specifications harder. The structural slop filter (marketing-cue density, keyword stuffing, bullet symmetry) also fully recovers (regret 0.250, success 0.575): the generator families leak templated structure even when their text is organic. Detection-style defenses, by contrast, are brittle in deployment: the perplexity filter, calibrated to near-perfect separation on held-out listings, collapses at high dose (success 0.200, *below* the unprotected 0.450) because its hard threshold drops the true product — whose generic text looks "predictable" — while a tail of synthetic pages survives. The matched random filter confirms the mechanism: dropping 30% of candidates uniformly *raises* the clean baseline (regret 0.172) yet does nothing against contamination (0.431) — filtering *amount* is not filtering *quality*. The prompt defense achieves an intermediate 57% regret recovery but changes behavior on the clean Web as well (a "skepticism tax"), so its net value is smaller than its high-dose delta suggests. Beyond the battery, four targeted experiments pin down *how* provenance information should be consumed and how robust the defenses are (all at ε=0.75, 40 paired tasks, seed-1 pools unless noted).

(i) Order, don't score. Re-implementing the provenance bit as a soft bonus $\lambda \cdot \mathbb{1}$[verified] added to the platform score fails for every weight within the rating scale — regret 0.507 (λ=0.5), 0.444 (1), 0.494 (2), 0.464 (4), 0.360 (6) — and only recovers at λ=8 (0.332, statistically tied with the hard re-ranker's 0.329). A bonus smaller than the typical gap between a manufactured 4.9-star rating and a genuine one is simply outbid: provenance must act as a *lexicographic* ordering constraint, not an additive score.

(ii) Forgery envelope. Letting the adversary forge the badge on s% of synthetic listings leaves recovery unchanged across the envelope: regret 0.313 (s=35), 0.332 (50), 0.325 (65) — all statistically

Table 4: Defense battery at $\varepsilon = 0.75$ (mix contamination, 40 paired tasks; `none@0` row = paired uncontaminated baseline). Recovery = regret excess over the uncontaminated baseline eliminated.

Defense	Success	Regret	Synth. bought	Recovery
none @ ε=0	0.525	0.338	0%	—
none @ ε=0.75	0.450	0.398	47.5%	0%
random (matched drop)	0.425	0.431	52.5%	−55%
prompt	0.500	0.364	37.5%	57%
detector-struct	0.575	0.250	45.0%	>100%
detector-text (ppl)	0.200	0.578	85.0%	−300%
provenance (noisy bit)	0.550	0.329	32.5%	≈100%
oracle (perfect bit)	0.750	0.171	17.5%	>100%

indistinguishable from the no-forgery 0.329 (Wilcoxon $p \geq 0.96$). The lone s=25% point scores *better* than baseline (0.222, $\Delta = -0.107$, $p = 0.012$); five trajectory replays attribute the gap to greedy-chain behavioral variance rather than a mechanism, and one favorable outlier among four tested rates is consistent with multiplicity. Because the badge derives from a provenance bit rather than listing text, a forger who *displays* the badge still loses: the re-ranker demotes every badge-carrying synthetic into the same tier, where the agent's specification check rejects it.

(iii) A naive defense backfires. Ranking candidates by rating× log(review count) — the most natural platform-side fix — is harmless on the clean Web (regret 0.256) but *amplifies* the attack: success 0.300, regret 0.507 (vs. 0.443 unprotected), and a synthetic listing bought in 67.5% of tasks. Manufactured social proof hijacks the ranking signal: defenses built on the same surface features the attacker optimizes are structurally complicit.

(iv) Stealth pools (arms race). The structural detector keys on generator template fingerprints (superlatives, bullet-length symmetry, `X:Y` spec lines, review boilerplate). Regenerating the pool with fingerprint-free prompts ("stealth" style: no marketing adjectives, naturally varied bullets, prose-form specifications, humanlike reviews) makes the attack itself slightly milder (unprotected regret 0.425 vs. 0.443 on the original pool) and partially degrades the detector: its regret recovery falls from 0.193 to 0.138 — a 28% loss, though the remaining recovery is still significant (paired Wilcoxon $p = 0.022$). The provenance badge is untouched by the stylistic shift: its recovery is 0.152 on the stealth pool versus 0.114 on the original one, exactly as expected for a defense that never reads the text. The arms race against content-based detection is thus real but gradual, while the provenance channel is structurally immune to it.

5.6 Backbone generality and scale

The dose–response replicates on Llama-3.1-8B (success 0.475 → 0.300 at $\varepsilon = 0.75$, −17.5 pp; regret 0.346 → 0.502, +45%; synthetic purchases 55%). Across scales, smaller agents are *at least* as susceptible — at $\varepsilon = 0.75$, Qwen2.5-3B buys synthetic listings in 70% of tasks (vs. 54% for 7B) and loses 27% more regret — but the success-level dose contrast requires minimal baseline competence, which Qwen2.5-1.5B lacks (success 0.300 at $\varepsilon = 0$, statistically unchanged under contamination). The vulnerability is not an artifact of one model's quirks. Summarizing the two competent backbones by a linear erosion coefficient (regret increase per unit contamination, κ from the endpoint regression), Qwen2.5-7B gives $\kappa = 0.242$ (95% CI $[0.129, 0.352]$) and Llama-3.1-8B $\kappa = 0.208$ (95% CI $[0.037, 0.374]$) — overlapping intervals, the same order of vulnerability — while the below-competence 1.5B backbone shows a floor effect (clean regret already 0.578) that makes the dose contrast uninterpretable.

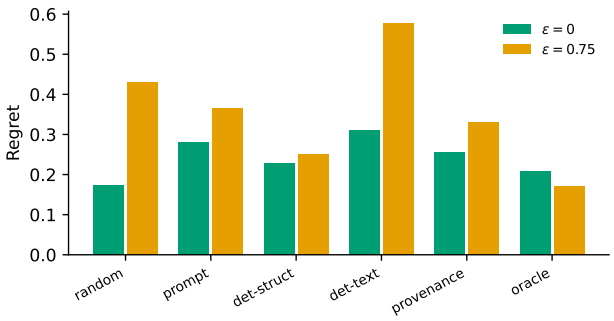


Figure 4: Regret at the uncontaminated baseline ($\varepsilon = 0$) and under $\varepsilon = 0.75$ mix contamination for each defense (40 paired tasks). Structural filtering and provenance badges restore uncontaminated-level regret; a hard perplexity threshold collapses below the unprotected condition; matched random dropping does not defend.

6 DISCUSSION

Preference, not perception. An earlier reading of our probe — which we report because it is the one a trajectory skim naturally produces — was that agents "cannot tell" clones from real products. The controlled decomposition falsifies that reading. At price parity the clone is *rejected* ($\sigma = 0.386$); at rating parity it is rejected identically (0.386); at both, it is rejected almost 3:1 (0.250) — and the two channels decompose additively, each worth roughly 14 percentage points. The dangerous form of ambient synthetic content is therefore not the attention-grabbing page nor the perceptually perfect forgery, but the *economically attractive counterfeit*: a listing that copies a real product's specifications and undercuts it with a discount and manufactured social proof. The pricing dose–response sharpens this into a threshold law — a 10% discount flips the majority, deeper discounts add nothing — which means the attack requires no optimization of the forgery itself, only access to the pricing surface. This recasts the threat model: attacks target the agent's *preference alignment* (what it trades off), not its perception (what it can detect), and defenses that shore up detection leave the actual channel open. It also explains why detector-style defenses based on surface statistics can only partially help: the clone's text statistics look organic because its text *is* organic, copied from a real product. The structural detector nevertheless performed on par with the provenance oracle in our seed-1 runs — and on fingerprint-free "stealth" pools it retains 72% of its recovery ($p = 0.022$) while the provenance badge is unaffected — but we caution that this robustness rests on generator fingerprinting and should be treated as arms-race-bounded.

The economics of silent erosion. Success rates stayed nearly flat while regret climbed and confidence rose. An operator watching task success — the metric most dashboards show — would see nothing. The losses are absorbed inside "successful" sessions as attribute mismatches and overpayment, which is precisely why an ambient contamination problem can grow for years without surfacing as an incident. The confidence inflation compounds it: agents err more while trusting themselves more, so downstream automation (reordering, auto-purchase budgets) inherits correlated, concealed losses. We argue this pattern — flat success, rising regret, rising confidence — should become the standard diagnostic triplet for evaluating agents on live-content distributions, in the same way calibration curves preceded accuracy as the trustworthy signal for classifiers under distribution shift.

Provenance as an interface contract. Our defense results argue for a specific division of labor — and for a specific *interface semantics*. Content-level detection asks every consumer to re-litigate every page's authenticity, and it fails exactly on the content that matters (near-copies of real pages). Provenance-weighted ranking instead asks the supply side for one bit — verifiable origin — and lets the agent treat its absence as a prior, not a verdict. Our λ-sweep adds a semantics-level finding: the bit must enter the ranking as an *ordering constraint*, not an additive bonus. Soft integration fails for every weight within the rating scale ($\lambda \leq 4$) because manufactured 4.9-star ratings outbid it, and only recovers once the provenance term dominates ($\lambda = 8$); a naive popularity-signal defense that consumes rating×review-count — precisely the features the attacker manufactures — actively amplifies the attack. An interface contract that exposes provenance should therefore expose it as a sortable field, and platforms should treat "provenance-blind" ranking signals as attack surface. Because the bit is cheap to spoof in the model but expensive to spoof in the world (a signature over seller identity and listing history), it converts an unbounded text-classification arms race into a bounded key-management problem — and our forgery envelope (unchanged through 65% badge forgery) shows the noisy-bit design tolerates adversarial display spoofing. The noise we inject (15% missing on real items, 15% present on synthetic ones) is an intentionally pessimistic model of real-world deployment, and recovery remains high; the remaining gap to the oracle is the price of noisy provenance, not of the idea.

Limitations. Our agents are 7–8B open-weight models with a scripted action interface; frontier models with stronger browsing may behave differently — though the channel decomposition suggests the vulnerability lives in preference alignment rather than perceptual limits, so capability alone may not close it; the erosion coefficient is statistically indistinguishable across our two 8B-class backbones (0.242 vs. 0.208, overlapping CIs). The mechanism decomposition and the pricing threshold law are measured on one backbone (Qwen2.5-7B) with one generator pair; both are pre-registered comparisons released with the paper, and we flag cross-generator replication as the natural next test. The contamination families are generated by models of the same size class as the agents; industrial content farms use larger generators, which could tune prices and ratings more aggressively, but our threshold law suggests any discount at all suffices. WebShop instructions are variant-level and sometimes weakly aligned with the base listing text, an inherent property of the benchmark that we handle with the official best-variant scoring; this makes absolute success numbers benchmark-specific but does not interact with the contamination manipulation, which is randomized within task. Our dose–response uses a single item pool per task; we mitigate pool-specific artifacts by keeping pools fixed across ε (paired design), by replicating the defense arms on an independent second generation pool, and by releasing all pools and seeds. Two secondary findings are reported at reduced strength after replication: pooled over two generation pools, the geo-family regret excess is directional only (CI crosses zero), and the single favorable point in the forgery envelope (s=25%) is adjudicated as behavioral variance rather than mechanism. Provenance availability in the wild is an empirical question our 85/15 noise model approximates crudely; measuring real provenance coverage on major marketplaces is a natural next step.

Outlook. If agentic browsing becomes a dominant traffic source, the incentive to optimize *for agents* will exceed the incentive to optimize for human attention, because agents can be measured and A/B-tested at scale. The question is not whether the Web's supply side adapts to machine readers — it already is — but whether machine readers will be able to tell what adapted. Ambient contamination is the benign end of that adaptation spectrum; deliberate manipulation is the malign end, and both flow through the same missing infrastructure: content whose origin a buyer can verify.

7 CONCLUSION

We asked whether the synthetic content already saturating the Web degrades the decisions of the agents that browse it, and answered with a controlled, paired, dose-controlled protocol on official benchmark artifacts. Agents are not naive, and — our central revised finding — they are not fooled either: channel decomposition shows they reject clones once discounts and manufactured ratings are removed. What fails is their *preference alignment*: a ten-percent discount and a manufactured 4.9-star rating are sufficient to flip the majority of decisions, so ambient content farms do not need to defeat the agent's perception, only to outbid it. Consequently regret inflates with contamination dose while success metrics and agent confidence tell a reassuring story. A single noisy provenance bit recovers the full loss — but only when the interface consumes it as an ordering constraint rather than a score, a semantics we argue should become a first-class contract between the Web and the agents acting on it. We release the SynthFlood protocol, contamination pools, and all trajectory logs to make agent-Web robustness a measurable property rather than an anecdote.

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A EXPERIMENTAL SETUP AND ADDITIONAL RESULTS

A.1 Models and serving

All agents and generators run locally via vLLM 0.19.0 (OpenAI-compatible API, `--enforce-eager`, temperature 0 for agents and the judge, temperature 0.7 with a fixed seed for the generator) on two RTX 4090 GPUs (24 GB): Qwen2.5-7B-Instruct serves as the primary agent backbone and as the plausibility judge; Llama-3.1-8B-Instruct serves as the contamination generator and as the secondary agent backbone. Qwen2.5-3B/1.5B are used in the scale ablation. Every LLM call is logged with its token counts; agents cap episodes at ten steps, forced-choice probes at six.

A.2 Official artifacts

We use WebShop's official `items_shuffle.json` (1,181,430 products after the official key-cleaning and deduplication), `items_ins_v2.json`, and `items_human_ins.json` (10,136 products with human instructions), with the benchmark's own loader, goal builder (`get_human_goals`, including its price-upper sampling), and reward function applied unmodified. Our loader is a verbatim port with paths parametrized; the only semantic addition is scoring a chosen listing through its best-matching variant, since the Select protocol has no variant-selection step.

A.3 Contamination pool

The pool covers 100 tasks (the official goal list under seed 2027), three families, four listings per family per task (up to 480 per task set), with a plausibility floor at judge score 4 (regenerate once, then drop). Synthetic economics are assigned programmatically: price = gold price $\times$ U(0.75, 0.95) (deterministic per listing), rating U(4.6, 4.9), review counts in $[400, 3000)$, and reviews in packs of 20 (reviews family) or 6 (others). An archived pre-fix pool with a \$50 price anchor is released as an ablation.

A.4 Gate outcomes

P0: success 0.65 at $\varepsilon = 0$, zero malformed actions, 18/20 tasks exactly reproducible under greedy decoding (two flips attributable to continuous-batching numerics). P1 (final, corrected pool): $\sigma = 0.377$ [0.324, 0.433] over 300 duels; per family: clone 0.523, geo 0.351, reviews 0.287. P2 (gate run, 40 paired tasks): success 0.525 $\rightarrow$ 0.450, regret 0.338 $\rightarrow$ 0.398, confidence 0.783 $\rightarrow$ 0.862 at $\varepsilon = 0.75$ versus 0. P4A (main curve, N=100 per dose): success 0.590 $\rightarrow$ 0.360 (McNemar $p = 1.3 \times 10^{-4}$), regret 0.262 $\rightarrow$ 0.443 (paired $\Delta = +0.181$, CI $[+0.095, +0.264]$), confidence 0.822 $\rightarrow$ 0.875, ECE 0.249 $\rightarrow$ 0.496; P4E replicates the endpoints under a second contamination seed (pooled McNemar $p = 1.2 \times 10^{-8}$). Two earlier P1/P2 runs were invalidated by a rendering defect (machine-readable review dictionaries leaking into rendered pages) and a price-anchoring bug; both are documented with their data in the repository.

A.5 Power and multiple comparisons

The paired regret difference at $\varepsilon = 0.75$ has task-level SD ≈ 0.43; at $N = 100$ per dose the endpoint McNemar test detects the observed success drop ($p = 1.3 \times 10^{-4}$), and pooling the second seed tightens it further ($p = 1.2 \times 10^{-8}$); the paired regret difference is significant with a bootstrap CI excluding zero. Intermediate doses are reported as effect sizes with cluster-bootstrap CIs rather than pairwise p-values. We do not over-claim single small deltas; the interpretation rests on the joint pattern across outcomes (success, regret, purchase share, confidence, calibration) and its replication across families, backbones, scales, and defenses.